



AI GROUP

Interpretable Learning of Orbital Dynamics Perturbations using Kolmogorov–Arnold Networks

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March 2026

1 Profile of the Team Leader(s) & Expected Team Composition

The project is co-led by experts with complementary backgrounds in machine learning, scientific computing and applied physics. The team combines expertise in deep learning frameworks (PyTorch), data-driven modeling, and simulation-based pipelines.

The project aims to recruit a multidisciplinary team of 4–6 participants with expertise across artificial intelligence, physics and software engineering. We seek individuals with experience in one or more of the following areas:

- Machine Learning / AI: Experience with neural networks, PyTorch, and model evaluation.
- Physics / Aerospace / Applied Mathematics: Knowledge of classical mechanics, orbital dynamics, or dynamical systems.
- Software / Simulation / Data Engineering: Experience with numerical simulation, Python, and data pipelines.

Participants are expected to work across domains enabling flexible collaboration and a tight integration of data-driven and physics-based approaches. The team will work collaboratively, with overlapping responsibilities, ensuring both rapid prototyping and scientific rigor.

2 Abstract

This project investigates the use of Kolmogorov–Arnold Networks (KANs) for learning physical laws governing orbital motion under perturbations in a way that is both accurate (competitive with high-fidelity numerical models) and interpretable (decomposable into physically meaningful components).

While classical Newtonian gravity accurately describes ideal two-body motion, real-world satellite trajectories are affected by complex perturbations such as atmospheric drag and Earth’s oblateness (J2 effect).

We propose a hybrid modeling framework combining known physical laws with a data-driven KAN component. Using simulated orbital trajectories, the KAN will learn the residual acceleration beyond Newtonian gravity. The goal is not only to achieve accurate predictions but also to extract interpretable relationships between state variables (e.g., altitude, velocity) and perturbation effects.

The project will deliver a proof-of-concept hybrid orbital propagator, comparative evaluation against standard neural networks and other mainstream methods, and an analysis of the interpretability of learned physical relations. This work contributes to the growing field of scientific machine learning by demonstrating how KANs can bridge physics-based modeling and interpretable AI.

3 Background Information & Problem Statement

3.1 Orbital Mechanics

Orbital mechanics is traditionally modeled using well-established physical laws derived from Newtonian mechanics. The fundamental equation governing gravitational motion is:

$$a = -\mu \frac{\mathbf{r}}{|\mathbf{r}|^3} \quad (1)$$

where $\mu = G M$ with G , the gravitational constant and M , the mass of the body producing the gravitational field.

However, real-world orbital trajectories deviate from this ideal due to perturbations such as:

- atmospheric drag (dominant in low Earth orbit) [1]
- Earth’s oblateness (J2 effect) [2]
- solar radiation pressure (effect of photons momentum on satellite surface) [3]
- third-body gravitational influences.

These perturbations are often modeled using approximations or simplified analytical formulations, which may not fully capture complex interactions across different regimes.

3.2 Physics Informed Neural Networks and Simulations

Physics-informed neural networks (PINNs) [4] provide a framework for incorporating physical laws, expressed as differential equations, directly into neural network training. By enforcing equations as soft constraints via automatic differentiation, PINNs enable learning of dynamical systems and have been widely applied to forward and inverse problems, system identification, and surrogate modeling. In this sense, PINNs establish a standard paradigm for combining deep learning with numerical simulation, typically relying on multilayer perceptrons (MLPs) as the basis function approximators.

Despite their flexibility, PINNs exhibit several well-documented limitations, including training instability, spectral bias, and difficulties in handling multiscale or stiff systems [5]. These challenges motivate a large body of work aimed at improving both the learning framework and the underlying architectures. Operator-learning approaches such as DeepONet [6] and Fourier Neural Operators [7] address scalability by learning solution operators. Also, recent research explores alternative function representations, including architectures such as KANs, which replace fixed activations with adaptive functional components. Surveys of the field [8, 9] highlight that improving expressivity, stability, and efficiency of neural representations remains a central challenge in physics-informed simulation.

3.3 KANs

Kolmogorov-Arnold Networks (KANs) [10] are a neural architecture motivated by the Kolmogorov–Arnold representation theorem, and proposed as an alternative to the standard multilayer perceptron (MLPs). Their key design replaces fixed activation functions with learnable univariate functions defined on edges, typically parameterized via splines. This formulation improves interpretability and has shown strong performance in scientific regression tasks, while enabling direct inspection of learned functional relationships.

KANs have recently been explored in scientific machine learning, particularly for differential equations and physics-informed modeling. Extensions such as Kolmogorov-Arnold informed neural networks (KINNs) [11], PIKAN [12] and KAN-based neural ODEs [13] demonstrate competitive or improved performance on ODE/PDE problems, although challenges remain in terms of stability, efficiency, and scalability. Comparative studies [14] and recent surveys [15, 16] highlight these limitations and motivate a growing body of work on architectural variants, including alternative basis parameterizations (e.g., polynomial or Fourier-based KANs).

3.4 Research Question

The key problem addressed in this project is:

Can Kolmogorov-Arnold Networks learn and interpret nonlinear perturbation forces in orbital dynamics while preserving physical consistency through a hybrid modeling approach?

This project is positioned at the intersection of:

- physics-informed machine learning
- interpretable AI
- astrodynamics modeling.

Orbital mechanics provides a challenge due to its complex, nonlinear, and multiscale dynamics. Addressing this problem is expected to yield insights useful in other domains involving structured physical systems, such as traffic modeling, aeronautical design, and broader simulation tasks where enhanced physics-based models can improve predictive performance.

4 Project Objectives & Concrete Implementation

This project aims to develop a hybrid orbital dynamics model that integrates analytical Newtonian mechanics with data-driven perturbation modeling using Kolmogorov–Arnold Networks. The goal is to improve predictive accuracy while preserving, or enhancing, interpretability and physical consistency.

4.1 Objectives

1. Develop a hybrid Newtonian-KAN orbital propagation framework.
2. Learn key perturbative forces (e.g., atmospheric drag, J_2 effects) from simulated orbital data.
3. Detect and characterize unmodeled or unaccounted dynamics in orbital motion.
4. Assess the interpretability of learned representations and their alignment with known physical laws.
5. Benchmark KAN performance against standard neural network architectures (e.g., MLP-based).

4.2 Expected Deliverables

1. **Hybrid Orbital Propagation Model:** a validated implementation combining analytical gravitational dynamics with KAN-based learned for perturbation forces.
2. **Simulation Pipeline and Dataset:** a fully reproducible pipeline for synthetic data generation, including configurable orbital regimes, perturbation models, and documented parameter settings.
3. **Experimental Framework:** a structured and documented codebase including
 - environment setup and dependencies
 - training and evaluation scripts
 - trained model checkpoints
 - extensibility guidelines.
4. **Evaluation and Interpretability Report:** a comprehensive analysis including

- **Quantitative evaluation:** comparison of KAN and baseline models in terms of prediction accuracy, generalization across orbital regimes, and computational efficiency.
 - **Interpretability analysis:** examination of learned functional relationships, including visualization of univariate components, identification of dependencies between state variables and perturbations, and assessment of physical plausibility.
 - **Dissemination:** preparation of results for submission to a peer-reviewed venue, subject to outcome quality.
5. **Final Report and Presentation:** a detailed report summarizing methodology, experimental results, and key findings, accompanied by a presentation.

4.3 Concrete Implementation

4.3.1 Simulation Pipeline

Synthetic orbital trajectories will be generated using a high-fidelity propagator. The simulation will cover diverse regimes, including:

- circular and elliptical orbits
- altitude ranges spanning LEO, MEO, and GEO
- varying spacecraft characteristics (e.g., mass, drag coefficient, area-to-mass ratio).

4.3.2 Dataset Construction

The dataset will consist of:

- **Inputs:**
 - position (x, y, z) and velocity (v_x, v_y, v_z)
 - derived features such as altitude, speed, radial distance r , and normalized components (e.g., z/r).
- **Targets:**
 - total acceleration, or
 - residual perturbation acceleration (preferred for hybrid modeling).

4.3.3 Model Development

Three model classes will be implemented and compared:

- a Kolmogorov–Arnold Network (KAN)
- a baseline multilayer perceptron (MLP)
- a well known approach [6] for SDE and ODE computing and modeling.

Models will be trained to predict acceleration or perturbation components from state variables.

4.3.4 Hybrid Model Formulation

The hybrid dynamics model combines:

- analytical Newtonian gravitational acceleration

- a learned correction term produced by the KAN.

$$\mathbf{a}_{tot} = \mathbf{a}_{Newton} + \mathbf{a}_{KAN} \quad (2)$$

4.3.5 Evaluation Protocol

Model performance will be assessed using:

- **Prediction accuracy:** MSE and MAE on acceleration estimates.
- **Physical consistency:** long-horizon trajectory rollouts and deviation from ground-truth simulations.
- **Comparative analysis:** performance comparison between KAN and the other analytical baselines.

4.3.6 Interpretability Analysis

The interpretability of the KAN model will be evaluated through:

- visualization of learned univariate functions,
- analysis of relationships between physical variables (e.g., altitude vs drag, velocity vs drag, geometric dependence of J_2)
- identification of nonlinear behaviors and regime transitions
- verification of consistency with established physical laws.

5 Detailed Work Plan

The project is structured over a two-week period, with parallel work streams across team members to ensure efficient execution.

5.1 Week 1 — Foundations and Baselines

Day 1: Setup and Planning

- Define system architecture and assign responsibilities.
- Set up the code repository and computational environment.
- Implement a two-body orbital simulator using libraries such as *poliastro*, *boinor* and *orbitalpy*.

Days 2–3: Data Generation and Training Strategy

- Generate synthetic trajectory datasets, including Newtonian dynamics and controlled perturbations.
- Validate simulation outputs for physical consistency.
- Define the training strategy, including progressive incorporation of perturbation effects.

Days 4–5: Initial Training and Baselines

- Train baseline models:
 - multilayer perceptron (MLP)
 - PINN based solver
 - initial KAN model (Newtonian-only setting).
- Evaluate baseline performance and convergence behavior.

5.2 Week 2 — Hybrid Modeling and Analysis

Days 6–7: Residual Learning

- Compute perturbation residuals from simulated data.
- Train the KAN model to predict residual acceleration.
- Analyze and visualize learned functional components.

Day 8: Hybrid Model Integration

- Integrate analytical Newtonian dynamics with the learned KAN correction.
- Perform trajectory rollout experiments to assess stability and consistency.

Day 9-10: Evaluation & Finalization

- Compare models in terms of prediction accuracy and trajectory stability.
- Evaluate generalization across orbital regimes.
- Extract and interpret physically meaningful patterns from the learned models.
- Prepare the final report, presentation, and demonstration.
- Consolidate code, documentation, and experimental results.

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